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Don L. Bunker and W. L. Hase

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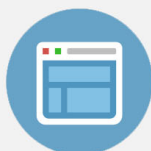
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On non-RRKM unimolecular kinetics: Molecules in general, and CH_3NC in particular*

Don L. Bunker and W. L. Hase

Department of Chemistry, University of California, Irvine, California 92664

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Monte Carlo rate constants for model CH_3NC isomerization, determined at 200, 100, and 70 kcal/mole, disagree with theoretical predictions. Also, three different approximate methods of generating initial conditions at 200 kcal lead to divergent results. The molecule does not appear to us to obey the random lifetime assumption of conventional unimolecular rate theory at any of these energies. A discussion is given of the systematics of this kind of effect, and comments are made on the relationship between our results and those obtained in the laboratory.

I. THEORETICAL PROLOGUE: WHAT TO EXPECT FROM A UNIMOLECULAR REACTION

A. Introduction

One of the best reasons for making a simple theoretical model of chemical behavior is merely that it can be done. A recognizable physical assumption, leading by tractable mathematics to definite predictions, provides a valuable guide, regardless of whether most reactions conform to it.

The RRKM¹ theory of unimolecular reactions is an example of such a model. It appears to explain a very large amount of chemistry. But even if it were not so successful, it would still be of central interest, because its main assumption is the only manageable one that can be made. Its converse implies that we cannot calculate reaction rates by any simple manipulation of the structural properties of cold reactant and critically activated molecules. Any treatment of non-RRKM kinetics must be much more complicated than the currently standard theory.

Since that is what we must discuss in this paper, we need to supply at least a rudimentary descriptive pathology of non-RRKM behavior. A bibliography is provided,²⁻⁹ and the notation has been chosen to conform as much as possible to that of Robinson and Holbrook.²

The basic assumption of RRKM theory is that isolated molecules behave *as if* all their accessible states were occupied in random order. In Fig. 1(a) we have illustrated this schematically, showing random transitions among states at some energy high enough for eventual reaction (towards the right). This assumption implies that there will be a random lifetime distribution

$$P(\tau) = k_a \exp(-k_a \tau), \quad (1)$$

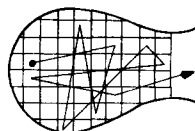
giving equal probability during any time interval for reaction to occur.

The "as if" wording is important. In reality,

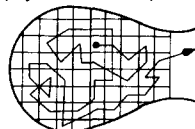
transitions between states are not random; some are more probable than others. In the limit of high energy and heavy atoms, only one transition is possible and we have a classical trajectory. Thus we have to rely on the molecular motion to be disorderly enough for the RRKM assumption to be mimicked, as crudely illustrated in Fig. 1(b).

In practice we never quite attain this ideal. To apply RRKM theory, we must prescribe the manner

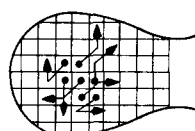
(a) RRKM model



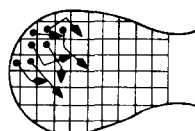
(b) physical counterpart



(c) collisional state selection



(d) chemical activation



(e) intrinsically non-RRKM

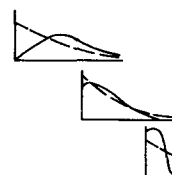
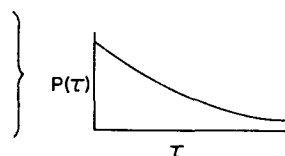
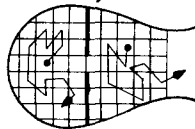


FIG. 1. Relation of state occupation (schematically shown at constant energy) to lifetime distribution for the RRKM theory and for various actual situations. Dashed lines in lifetime distributions indicate RRKM behavior.

in which the energy needed for reaction is supplied. The most usual procedure for thermal reactions, implicit in Figs. 1(a) and 1(b), is the *detailed strong collision assumption*. This says that after an activating collision, any energy is likely in proportion to its equilibrium probability and that the states at that energy are all equally probable. It is the deficiencies of the second part of this assumption that we have to explore.

It has been pointed out previously⁸ that thermal collisions surely obey at least an approximate Franck-Condon principle. Direct collisional preparation of a state having a large extension of some bond, in the direction of product formation, is suggested to be an implausible event. We might therefore expect some shortage of lifetimes less than $\frac{1}{4}$ of a vibration period, as illustrated in Fig. 1(c). This effect would be important only at extremely high pressures, and will not be further discussed here (everywhere, dashed curves show RRKM behavior for comparison).

A more interesting situation is that in which detailed strong collision conditions are intentionally avoided. This is done in a large number of experiments, prominently including those based on chemical activation. No matter how unusual the pattern of initial energization, the RRKM assumption 1(a) would render it equivalent to a random one in a negligibly short time. How well will the actual motion 1(b) approach this ideal? Three of the possibilities are represented by the lifetime distributions in Fig. 1(d). Deposition of the energy in favorable or unfavorable locations in the molecule might depress or enhance the probability of a short lifetime with respect to reaction. Intermediate situations or strong internal coupling would yield a $P(\tau)$ not very different from a random one. The latter alternative seems to be consistent with most of the experimental observations that have been made.

We have been describing *apparent* non-RRKM behavior arising from state selection. An entirely distinct possibility is an *intrinsically* non-RRKM molecule. By this we mean one that would have a nonrandom $P(\tau)$ even if the detailed strong collision assumption were exactly true. This could arise if transitions between two (or more) groups of states were less probable than those leading to products. A group of states not well coupled to the product channel, illustrated at the left in Fig. 1(e), would introduce a component of long lifetimes into $P(\tau)$, with qualitative results as shown in the drawing.

Curves like the $P(\tau)$ in Fig. 1(e) must descend continuously. The easiest way to see this is to think of the lifetimes associated both with the forward (τ) and backward (τ_b) directions in time. The

"gap length" l is $\tau + \tau_b$. All three of these must have random distributions if any do, and it is possible both to formulate the theory and discuss its possible failure⁴ in terms of l . Gap and lifetime distributions are related by

$$P(\tau) = \int_{\tau}^{\infty} P(t) dt. \quad (2)$$

There is one lifetime of length τ for every gap with $l > \tau$. So $P(\tau)$ can have a maximum other than at $\tau = 0$ only if $P(t) < 0$ for some t , which is impossible.

B. Pressure Dependences

In preparation for Sec. II, we discuss the possible ways in which reaction rates might be found to depend on the frequency of molecular collisions. We wish to distinguish four cases: (1) No significant deviations from the RRKM assumption under the experimental conditions—Fig. 1(b), 1(c), and the middle $P(\tau)$ of 1(d); (2) apparent non-RRKM features arising only from state selection—the other $P(\tau)$ of 1(d); (3) intrinsically non-RRKM—1(e); (4) both (2) and (3). The discussion will pertain to a single energy. The effect of averaging over these energies would be a further blurring of distinctions that we expect to show are already rather unclear. Strong collisions are assumed in a non-detailed way. We propose only to say that an energized molecule that suffers a collision will be deactivated. Failure of this would also introduce further indistinctness among the four cases.

The probability that a molecule avoids a collision for τ seconds is

$$W(\tau) = \exp(-k_2[M]\tau), \quad (3)$$

in which k_2 is a bimolecular rate constant for deactivation of reactant molecules by any collider M. Similarly, the rate of collisional preparation is $k_1[M]$. The over-all rate for a particular energy is then

$$k(E) = k_1[M] \int_0^{\infty} W(\tau) P(\tau) d\tau. \quad (4)$$

The integrand expresses the joint probability that the state formed has lifetime τ and that there is no deactivating collision within this time. The integral covers all molecular states at energy E .

Case (1). For RRKM kinetics, substitution of Eqs. (1) and (3) in (4) yields the familiar form

$$k(E) = \frac{k_1[M]}{(1 + k_2[M]/k_a)}. \quad (5)$$

At high pressures

$$k(E) = k_a(k_1/k_2), \quad (6)$$

and k_a can be evaluated in detail by statistical procedures.²

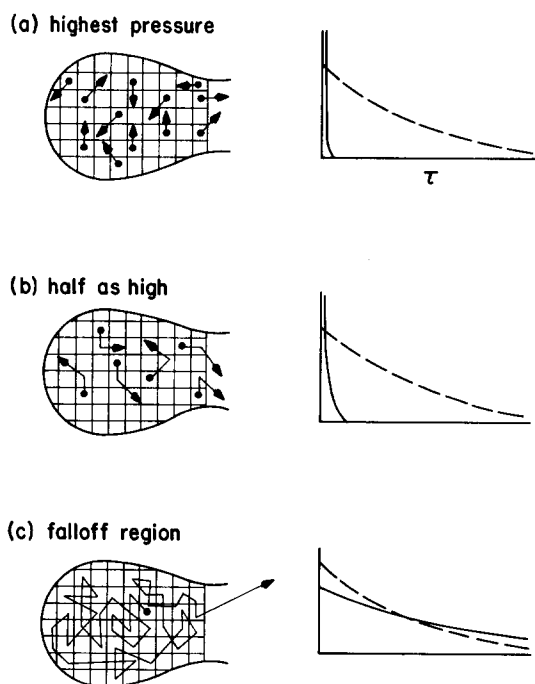


FIG. 2. Pressure dependence of an RRKM molecule. Dashed lines are random lifetime distributions and solid lines distributions of time until next collision, $e^{-k_2[M]\tau}$.

In Fig. 2 we have illustrated the overlap of the two random distributions $P(\tau)$ and $W(\tau)$ to produce the characteristic "falloff" behavior of unimolecular reactions (e.g., dashed curves in Fig. 3). If we imagine a pressure so high that only one intramolecular transition has time to occur between collisions, it is very clear that the reaction rate is determined by the statistical weight of states adjoining the critical configuration, as in Fig. 2(a). In Fig. 2(b) the pressure has been halved. In the same time interval as for Fig. 2(a), one-half as many molecules live twice as long and have twice the probability of reacting, so the net rate is the same as before. It begins to fall when there is time between collisions for most of the states to be occupied, so that reaction would be expected regardless of statistical weight considerations ($k_a \cong k_2[M]$). This is not the standard description of what happens, but is equivalent to it.

Case (2). If there are significant effects of state selection, the integral in Eq. (4) cannot be evaluated without further information. Some qualitative possibilities are illustrated in Fig. 3. In 3(a) we have superposed, at the left, Eq. (3) and the two unusual $P(\tau)$ from Fig. 1(d), and shown the expected kind of deviations from the RRKM form of $k(E)$ vs pressure.

A specific example is of interest. The molecule HBC (2, 2-dideutero-2', 2'-diprotiohexafluorobicy-

clopropyl) has been shown by Rynbrandt and Rabinovitch¹⁰ to have apparent non-RRKM behavior. Rates of CF_2 elimination appear to depend on which ring is initially energized. At sufficiently high pressure, the yield ratio for the two CF_2 elimination channels departs from unity.

For this effect to arise from state selection in exactly the way shown in Fig. 3(a) requires that the two deviations that contribute to the ratio have their onset in the same pressure region. It is by no means necessary for this to occur. Two other possibilities are represented in Figs. 3(b) and 3(c). The pressure at which the ratio first begins to rise gives information about the energy relaxation time between the rings if it is 3(a) or 3(b), but not 3(c). In general, the pressure dependence gives some kind of information, in the form of an integral over the lifetime distribution, but this is not necessarily of a simple nature. Applying the RRKM theory to part of the molecule, in order to clarify the situation, would implicitly reclassify the behavior as Case (4) rather than (2)—as is indeed at least somewhat suggested by the structure of HBC. In this paper we take no position on how molecules like

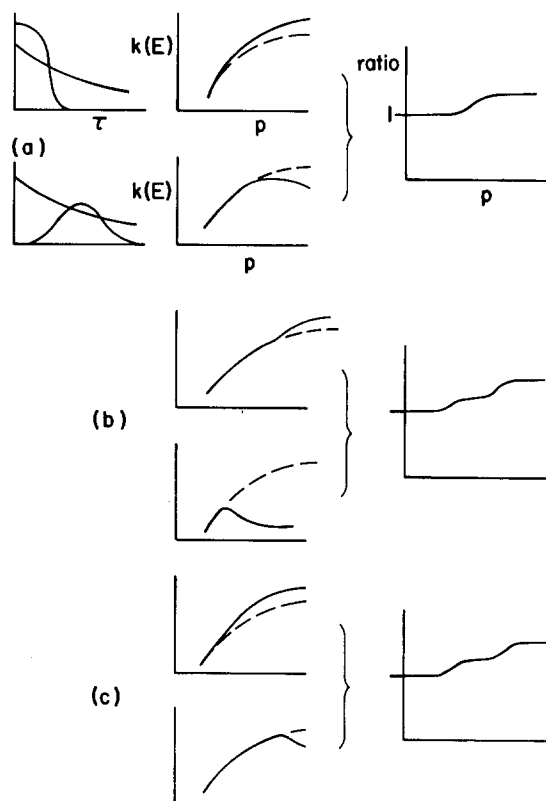


FIG. 3. Left: overlap of $e^{-k_2[M]\tau}$ with possible state-selected lifetime distributions. Center: resulting apparent deviations from RRKM (dashed) falloff. Right: ratio of $k(E)$ for the pairs of curves bracketed.

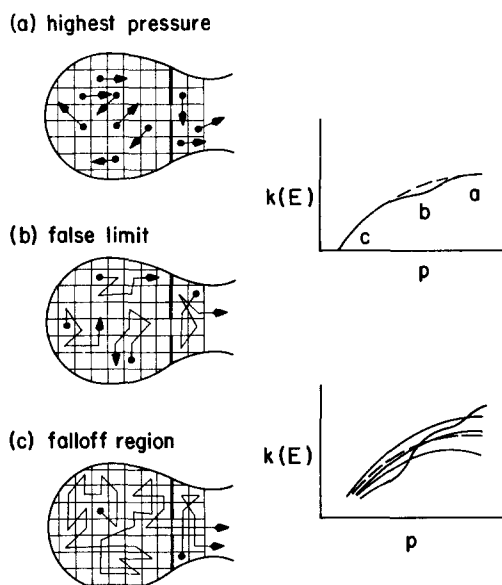


FIG. 4. Left: similar to Fig. 2 but for an intrinsically non-RRKM molecule. Upper right: corresponding regions of the falloff curve. Lower right: some possibilities if state selection is also present.

HBC are to be classified.¹¹

Case (3). Figure 4 illustrates what happens when there is intrinsic nonrandomness, but no state selection. The relatively severe case in which most of the states are badly coupled to the reaction is shown. Comparison of Figs. 4(a) and 2(a) establishes that at sufficiently high pressure the impaired internal coupling will have negligible effect, and RRKM behavior will be found. As usual, at low enough pressure the reaction rate is the same as the collisional preparation rate. The deviations occur in the intermediate pressure range.

The non-RRKM false high pressure limit arises if the RRKM depletion of states near reaction has reached its low pressure regime, but the transition rate between the groups of states has not. This is the meaning of the remark by Rice³ that at very high pressures the molecule would behave as if it had fewer atoms than are actually present. It must be emphasized that of the three pressure ranges identified in Fig. 4 and keyed to the upper falloff drawing, 4(a) and most of 4(b) are almost never explored in the laboratory. A conventional experimental falloff curve offers no immediate clue to what type of high pressure limit it approaches.

Let us suppose that a non-RRKM lifetime distribution, as in Fig. 1(e), can be represented by a sum of exponentials. This will generally be the case if intramolecular relaxation can be described by a master equation formalism⁶:

$$P(\tau) = \sum_i c_i e^{-k_i \tau}, \quad (7)$$

$$\sum_i c_i = k_a \quad (8)$$

since RRKM behavior is obtained at truly high pressure, and

$$\sum_i (c_i/k_i) = 1 \quad (9)$$

since all states are ultimately reactive. From Eq. (4)

$$k(E) = k_1[M] \sum_i c_i / (k_i + \omega), \quad (10)$$

in which we have let $\omega = k_2[M]$ to avoid subscript duplication, and

$$\frac{k(E)}{k(E)_{\text{RRKM}}} = \frac{\omega + k_a}{k_a} \sum_i \frac{c_i}{\omega + k_i}. \quad (11)$$

This approaches 1 if either $\omega \rightarrow 0$ or $\omega \rightarrow \infty$. If the sum has only two terms, as in Fig. 4, and we use Eqs. (8) and (9) to eliminate the c_i , we obtain

$$\frac{k(E)}{k(E)_{\text{RRKM}}} = \frac{\omega + k_a}{k_a} \frac{k_I k_{II} + \omega k_a}{(\omega + k_I)(\omega + k_{II})}. \quad (12)$$

We have let $i = I, II$, again to avoid confusion with the k_2 used earlier, and omitted several stages of routine algebra.

In the severe case illustrated by Fig. 4,

$$c_{II}/k_{II} \gg c_I/k_I, \quad (13)$$

corresponding to the uneven state distribution shown. In the region of the false limit,

$$k_{II}, k_a \ll \omega \ll k_I. \quad (14)$$

Here $k_a \ll \omega$ because the pressure is higher than would be expected for RRKM falloff. Equation (12) approaches

$$k(E)/k(E)_{\text{RRKM}} = k_{II}/k_a + \omega/k_I. \quad (15)$$

Since the second term is small and Eqs. (13) and (9) imply $c_{II} \cong k_{II}$, the measured high pressure rate gives the intramolecular vibrational relaxation rate directly. Note that the c_i express the rates and the k_i the partitioning of states. If the inequality of Eq. (13) is replaced by an equality or reversed, the meaning of the measured limit becomes more complicated and finally RRKM behavior is approached.

The purpose of this treatment is only to show that in a molecule like CH_3NC with a relatively small reactive part, if non-RRKM behavior is present it will manifest itself at intermediate rather than high pressures. A theory of the effect would require a method of determining the c_i and k_i . One such procedure is used later in the paper.

Case (4). If there is state selection with an in-

trinsically non-RRKM molecule, almost any pressure dependence might be observed (lower drawing in Fig. 4). Selection of states of opposite degree of coupling to the reaction channel could raise or lower the apparent rate at high pressures, and any number of inflections might occur. The only certainty is that all the possibilities will merge on the low pressure range, which may or may not wholly include the experimental conditions.

C. Conclusions

The RRKM theory is a mathematical idealization and does not correspond to any actual molecular mechanics. Departure from it is always present. In the case of non-RRKM features arising from state selection, this is only a matter of degree, not of kind. A molecule may be free of intrinsically non-RRKM character. But if not, this too is a matter of degree, and the effects may be large or small.

When we say we have an RRKM reaction, *one* of the things we mean is that none of the deviant behavior discussed above has been detected. Of course we also demand that if we know the molecule's critical configuration, the RRKM statistical procedures will give k_a correctly; and that if we have to guess the critical configuration, unreasonable guesses will not be required.

It should be evident that aside from the statistical arguments, it is very difficult in a single experiment to establish non-RRKM kinetics, much less determine how it arises. Showing whether HBC is intrinsically non-RRKM, for example, would require additional thermal experiments of a kind whose successful execution is hard to imagine. In each case where the basic RRKM assumption is being examined, the molecule in question should be subjected to every feasible test—thermal studies, chemical activation of all possible kinds, high pressure measurements, satisfactory statistical rationalization of k_a , and any other procedure that may seem relevant. We submit the remainder of our paper in this spirit.

II. MONTE CARLO STUDY OF CH₃NC

We should state at the outset that we do not view CH₃NC as an ideal molecule for our kind of test of thermal unimolecular behavior. We became interested in it in connection with observations of secondary decomposition of hot-atom substitution products. Our earlier suggestion¹² from trajectory evidence of its thermal non-RRKM behavior was more or less an afterthought at the time. Subsequently the hot-atom part of the problem was solved by a different approach.¹³ This paper records our attempt to resolve the questions we raised about the conformity of thermal CH₃NC decomposition to the

RRKM theory. We intend to make a similar study on a better molecule. This means one that does not have unusual rotational effects,¹⁴ and that has been studied in the laboratory by chemical activation methods, so that the energy range of interest is not so inconveniently low as is the case here.

We present the design of our trajectory study, a description of our RRKM and other theoretical calculations, a relatively small number of results, and discussion, in that order.

A. Models Used in Trajectory Study

The simplest model used was

$$U = \sum_{i=1}^5 D_i \{1 - \exp[-\beta_i(r_i - r_i^0)]\}^2 + \frac{1}{2} k_\varphi \sum_{i=1}^3 (\varphi_i - \varphi_i^0)^2 + \frac{1}{2} k_\chi \sum_{i=1}^3 (\chi_i - \chi_i^0)^2 + \sum_{i=0}^5 \alpha_i \cos i\theta \quad (16)$$

It has five Morse bonds corresponding to the N=C group, the 3 C-H, and a linkage from the methyl C to the center of the NC group (not the N). Harmonic forces resist the distortion of the H-C-H angles χ and the H-C-(NC center) angles φ . The truncated Fourier cosine series expresses the variation of potential energy along the critical coordinate, which is the angle θ between NC and the bond from the NC center to the methyl C. This is the same critical coordinate as was proposed by van Dine and Hoffman.¹⁵ The Fourier coefficients were chosen to give a 37.8 kcal/mole potential barrier to isomerization to CH₃CN, with an exoergicity of 15.3 kcal/mole, and to approximately match spectroscopic bending frequencies¹⁶ when $\theta = 0^\circ$ and 180° .

In some calculations we added positive or negative bending anharmonicity by incorporating extra terms into Eq. (1):

$$f \left[\sum_{i=1}^3 (\varphi_i - \varphi_i^0)^4 + \sum_{i=1}^3 (\chi_i - \chi_i^0)^4 \right]. \quad (17)$$

In others we introduced a CH₃-(NC) torsional potential via repulsions between H's and the N and C of (NC), usually in conjunction with the use of Eq. (17). The extra torsional terms are

$$c \sum_{i=1}^3 [(r_{iN} - r_s)^2 + (r_{iC} - r_s)^2], \quad (18)$$

with the proviso that $r_{iN} > r_s$ or $r_{iC} > r_s$ leads to a zero contribution from the corresponding squared term. We had two forms of the torsional barrier—"high" (5 kcal/mole at $\theta = 40^\circ$, 3 kcal/mole at $\theta = 45^\circ$) and "low" (800 cal/mole at $\theta = 45^\circ$). These bracket a recent calculated value of 1.48 kcal/mole at $\theta = 45^\circ$, which also identifies the N-eclipsed conformation as the one of higher energy. The calculation is the SCF study of Schaefer and co-workers¹⁷

whom we wish to thank for their willingness to extend their original work in response to our request for information.

Values of all the parameters in Eqs. (16–18) are given in this paragraph. $D_{\text{CH}} = 102$, $D_{\text{NC}} = 212$, $D_{\text{C(NC)}} = 104.2$ kcal/mole; $r_{\text{CH}}^0 = 1.094$, $r_{\text{NC}}^0 = 1.16$, $r_{\text{C(NC)}}^0 = 2.02$ Å; $\beta_{\text{CH}} = 1.86$, $\beta_{\text{NC}} = 2.39$, $\beta_{\text{C(NC)}} = 2.27$ Å⁻¹; $k_\phi = 0.547$, $k_\chi = 0.459$ mdyn · Å/rad²; $\varphi_i^0 = \chi_i^0$ = tetrahedral angle (1.9148 rad); $\alpha_0 = 14.763$, $\alpha_1 = 9.433$, $\alpha_2 = -22.725$, $\alpha_3 = 0.065$, $\alpha_4 = 0.3120$, $\alpha_5 = 1.848$ kcal/mole. Positive bending anharmonicity: $f = 0.05$; negative, $f = -0.1$ mdyn · Å/rad². High torsional barrier, $c = 500$ kcal/mole, $r_\chi = 2.15$ Å, and α_i changed to 17.458, 9.974, -26.873, 0.667, 0.369, -1.595, respectively. Low barrier: $c = 54.5$ kcal/mole, $r_\chi = 2.2$ Å, α_i changed to 15.913, 9.433, -23.875, 0.065, 0.312, -1.848.

B. Trajectory Initial Conditions

Truly random thermal sampling of initial states, for a molecule of this size, is a problem of monumental difficulty. We have not solved it, nor do we think it deserves detailed attention when the number of sample points (< 6000) is so much less than the number of orthants¹⁸⁽²³⁰⁾ in the probability space. Instead, we used three approximate methods, each of which deviates somewhat from random in its own way. The results of doing this should be rather similar to collisional state selection, Fig. 1(c). Starting points will not be completely random within an orthant. Therefore, in a calculated distribution of molecular lifetimes with respect to isomerization, we must disregard a possible transient up to one-half vibration period wide, at the zero-lifetime edge of the distribution. We always extended our simulation of the molecular motion to long enough times for this to be done.

In describing our procedures, we shall have to refer to some of the standard technology of trajectory calculations. One of us has reviewed this material.¹⁹

1. Mode Sampling

This is a refinement of our original pseudothermal method.¹² An attempt is made to apportion energy on a reasonably accurate atom-by-atom and vibrational mode-by-mode basis. The procedure is cumbersome and is not preferred for routine use. In informal flowchart form, it works as follows. The description uses absolute Cartesian coordinates.

(1) Divide the energy equally between atoms (E_i), and for each atom, randomly between kinetic and potential (E_{ki} , E_{pi}).

(2) Choose the desired vector angular momentum L . If from a thermal distribution, it may be ob-

tained by standard sampling²⁰ procedures ($T = 500$ °K in our case).

(3) Assign trial velocity vectors $\mathbf{v}_i$ to atoms—for masses m_i , magnitudes are $(2E_{\text{ki}}/m_i)^{1/2}$, and senses are random.

(4) Calculate and subtract vectorially the center-of-mass velocity.

(5) If the new total kinetic energy differs from that of step (1) by more than a fixed small amount (usually 2 kcal/mole in our case), uniformly magnify or demagnify the $\mathbf{v}_i$ and return to step (4). Repeat until agreement is attained. Convergence is rapid.

(6) Calculate the spurious angular momentum L' , which arises in the same way as the spurious translation in step (4). Let

$$\omega_x = (L_x - L'_x)/I_x,$$

similarly for y and z , with I_i principal moments of inertia and ω_i components of ω . Add to each $\mathbf{v}_i$ the net rotational velocity $\omega \mathbf{r}_i$, in which $\mathbf{r}_i$ are vectors from the center of mass to each atom. The total kinetic energy now has to be checked again. Repeat from step (4) if necessary.

(7) In the remaining steps, potential energy is apportioned between groups of vibration modes. The four groups are the CNC bends, the CNC stretches, the HCH bends and CH₃ rocking modes, and the CH stretches, to which $\frac{1}{6}$, $\frac{1}{6}$, $\frac{5}{12}$ and $\frac{1}{4}$ of the energy is assigned *on the average*. In this step, construct a random four-membered distribution with this bias.

(8) Generate small trial displacements with magnitude $(0.25 E_{\text{pi}})^{1/2}$ and random sense. The 0.25 is arbitrary.

(9) Groups of coordinate displacements, loosely identified with mode groups, are to be sequentially magnified and demagnified until the distribution from step (7) is matched. The identifications are CNC bends with x , y , z coordinates of cyano C, N; CNC stretches with x , y , z coordinates of methyl C and z coordinates of cyano C, N; other bends and rocks with x , y , z of H's and CH stretches with x , y of H's. Convergence of this procedure requires up to 150 sweeps through the four mode groups. The initial condition selection is fully accomplished when it is attained. Conversion is then made from absolute coordinates to the relative Cartesians (methyl C suppressed) used in trajectory integration.

2. Orthant Sampling

Both the remaining sampling methods work in the molecular phase space. They share an essential

need—an efficient method of generating random vectors in spaces of arbitrary dimensionality. We are not aware of any previous algorithm for doing this, so ours is given in the Appendix. The coordinates are now those of the integration procedure, not absolute Cartesian.

We choose one of the 2^{30} orthants at random, and erect a unit random phase-space vector within it. Its length is then magnified until the correct energy is attained. The magnification is not uniform for coordinates, however; small linearized biases are introduced in order to make the more anharmonic direction of extension more probable. They are proportional to the maximum possible positive and negative bond extensions with the available energy.²¹ An approximately correct distribution of angular momenta is produced automatically by this procedure.²²

3. Progressive Sampling

This is a more sophisticated phase-space method with interesting properties. The starting point is any trajectory, initialized by one of the other methods. During integration, after about one-half of an average vibration period, a 30 vector of coordinates and momenta is saved. Call this vector $\mathbf{a}$.

The key to the method is the not easily visualized fact that in spaces of high dimensionality, randomly oriented vectors are almost certain to be nearly orthogonal. Thus if we now choose a random vector $\mathbf{r}$, values of the $(\mathbf{a}, \mathbf{r})$ angle will cluster just below 90° . We can now bisect this angle (easy in any dimensionality) and produce a new $\mathbf{r}$ whose angle with $\mathbf{a}$ is near 45° , and by repetition, 22.5° , 11.25° , etc. In practice, we specify a maximum angle ψ , and by repeated bisection displace $\mathbf{a}$ in a random direction through an angle between ψ and 0.5ψ . Scaling, like that used in orthant sampling, must then be applied. The next trajectory is then ready to start.

For $\psi = 90^\circ$, this method reduces to orthant sampling. For ψ very small, it starts trajectories in regions of phase space with high probability of traversal. Thermal sampling is approached, but at the expense of a requirement for very long selection chains in order to cover the whole space properly. For us, $\psi = 15^\circ$ was a reasonable choice. For $\psi = 3^\circ$, we found persistence effects—reactive trajectories failing to generate nonreactive ones, and vice versa—of greater length than our limited sample size would allow us to tolerate.

C. RRKM and Relaxational Calculations

For unimolecular rate constant calculations, we used our general purpose RRKM program package.²³ Since comparison is with classical trajec-

tory results, its state-counting procedures were not needed. Also, because this is a particularly uncomplicated isomerization, critical configuration selection by the minimum state density criterion is much less important than it usually is. External rotations were treated as fully adiabatic, and numerically generated anharmonicity corrections were present.

Parameters used in the RRKM calculation were obtained by straightforward vibrational and rotational analysis of our model molecule and its isocycles critical configuration. They are displayed in Table I. The presence or absence of bending anharmonicity and torsional barriers has little numerical effect. On the scale on which we will be examining discrepancies, other models^{12,24} of CH_3NC have quite similar RRKM rate constants.

Purely theoretical treatments of unimolecular reaction rates in the non-RRKM regime have appeared.⁷ These do not accept a level of molecular structural detail comparable to that of the RRKM theory, and we have not as yet tried to refine them. Rather, we attempted to use a semiempirical relaxation theory that was proposed by one of us to account for the results of previous trajectory studies on simpler RRKM and non-RRKM model molecules.⁶ The theory accepts essentially the same information as in Table I, plus anharmonicity data obtained from Morse and other parameters appearing in Eqs. (16)–(18).

In view of the very long extension of this theory from the conditions for which it was derived, we have made additional tests of its believability. We applied it to HBC with satisfactory results.²⁵ The predicted intramolecular relaxation rate is less than that of the reaction; agreement with its experimental magnitude is within a factor of 6. For *sec*-butyl, whose constant-rate regime extends to very high pressure,²⁶ we predicted a relaxation rate 17 times lower than would be required for an RRKM molecule. This is, however, on the assumption of strong collisions. The experimental collider was H_2 , which does not fulfil this description. Collision-induced intramolecular effects may alter the meaning of this test. We concluded that the predictions of the theory for CH_3NC will still be of some interest.

D. Results

Sample raw data appear in Fig. 5. The total energy is 200 kcal/mole and most trajectories are reactive. About five cycles of CNC bending motion, and many more than that for other modes, are present. Mode sampling was used. Besides producing the highest reaction rate, this method is characterized by a sharp positive short-time sam-

TABLE I. RRKM input data.

Mode	Frequency (cm ⁻¹)	
	Molecule	Critical configuration
CH stretch	3039 (2)	3039 (2)
CH stretch	2925	2925
N≡C stretch	2190	2190
CH ₃ deformation	1342 (2)	1342 (2)
CH ₃ deformation	1388	1388
CH ₃ rocking	941 (2)	941 (2)
C-N stretch	870	870
C-N≡C bend	444	Free rotor ($I_R = 2.34 \text{ amu} \cdot \text{\AA}^2$) Critical coordinate
C-N≡C bend	444	
Moments of inertia (amu · Å ²)	50.25	52.30
	50.25	43.62
	3.20	11.89
Symmetry	$\sigma = 3$	$\sigma = 3$

pling transient, as discussed earlier. Separation of the transient from the descending exponential required by RRKM theory can easily be done by eye. In RRKM theory the decay constant of the exponential should be the same as its zero-lifetime extrapolated intercept, as prescribed by Eq. (1). For our purposes, an accurate enough test of this is comparison of the total number of reactive events having lifetimes greater than about 10^{-13} sec with the number predicted by the RRKM calculation. The ratio of these can usually be equated to that of measured to predicted rate constants.

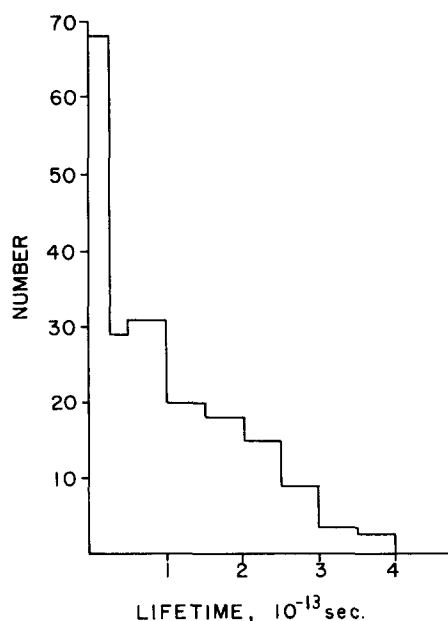


FIG. 5. Computed lifetime distribution with mode sampling at 200 kcal/mole. Other sampling methods yield many fewer events.

The theoretical and Monte Carlo rate constants are displayed in Fig. 6. Error ranges are based on the expected statistical fluctuation of the number of reactive trajectories. They appear unsymmetric because of the logarithmic ordinate. Points with downward arrows are upper limits. They rep-

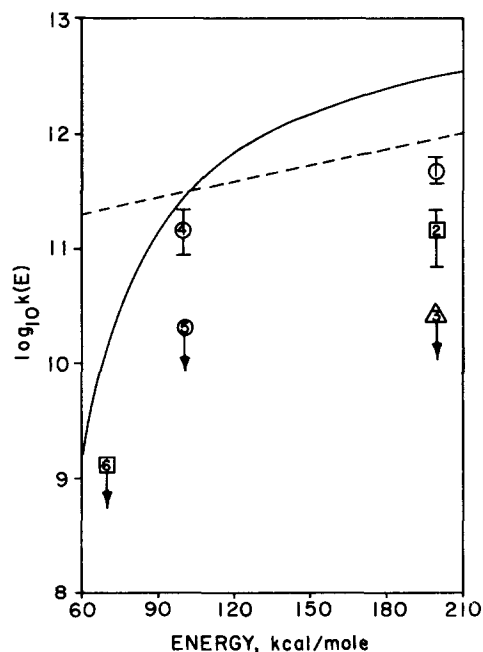


FIG. 6. Results: $\log$ (unimolecular rate constant) vs energy. Solid line: RRKM prediction. Dashed line: relaxation theory prediction. Circles: mode sampling. Squares: progressive sampling. Triangle: orthant sampling. Points 4 and 6 have positive bending anharmonicity and low torsional barrier. Others have neither, but at 200 kcal/mole, calculations with higher barrier and anharmonicity coincide with those shown, and at 100 kcal/mole, progressive sampling, low barrier, and anharmonicity produce the same result as point 5.

resent situations where *no* isomerizations were observed when a substantial number were predicted.

Consider first the relationship of the points at 200 kcal/mole to the RRKM curve (solid line). The three points represent the three different sampling methods, and *each is independent* of the molecular model used. The points match neither each other nor the results of the RRKM calculation.

According to the relaxation theory, we should expect a deviation from the RRKM curve, in the same direction as was found. The dashed line gives the theoretical intramolecular vibrational relaxation rate, which is practically independent of molecular model. RRKM behavior would be expected wherever the relaxation curve is the higher of the two—up to 100 kcal/mole in this example. This is much larger than the average energy of thermally isomerizing CH_3NC but would not be inaccessible in a typical chemical activation experiment, if one could be done. The relaxation theory in its present state has no provision for different patterns of initial energization; we are entitled merely to anticipate some agreement with the general range of the three 200 kcal/mole Monte Carlo points.

Calculations at 200 kcal/mole are fairly cheap. If the energy is lowered the cost rises rapidly, because of the large sample needed to produce a reasonable number of isomerizations.²⁷ We were able to make three calculations at 100 kcal/mole, near the crossing point of the theoretical curves. The results of these vary somewhat with molecular model. A calculation with the low torsional barrier and positive bending anharmonicity comes moderately close to agreement with both theories. One without these refinements, but with the same (mode) sampling, produced no reactions. Neither did progressive sampling with positive anharmonicity and low torsional barrier.

In view of this ambiguous outcome, we secured one more point at 70 kcal/mole, well below the crossing point of the two theoretical predictions. Progressive sampling, positive anharmonicity, and low torsional terms were used. This is our largest sample, 5050 trajectories, many times more than all the rest of the work combined. We expected 10 ± 3 isomerizations and observed none. One reaction event (the upper limit shown) would represent an order of magnitude downward deviation from the RRKM prediction. The chance that this result is an accident is about 1%. In the progressive sampling, we restarted the procedure every 50 or 100 trajectories, using equal numbers of reactive and nonreactive vectors scaled down from mode sampling results at high energies. This was

to ensure that any persistence bias would favor reaction. Mode sampling was avoided at this energy because it takes too long.

Our data are rather sparse, but we certainly view the contents of Fig. 6 as generally unsatisfactory agreement, considering the excellent results that this kind of test has yielded on other occasions.^{6,28} Since laboratory experimentalists have concluded that CH_3NC is a superior RRKM molecule, we now must carefully examine the relationship between our work and theirs.

E. Discussion

We have shown that a series of mathematical models of the potential energy for CH_3NC does not lead by classical mechanics to RRKM behavior. Here we discuss the obstacles to transferring this conclusion to real CH_3NC , in order of what seems to us their ascending seriousness.

Are our models sufficiently detailed? We think they are, as shown by the relative insensitivity of their inability to produce RRKM behavior to their properties, and by the lack of any comprehensible order of variance at 100 kcal. The SCF calculations also support the idea that our intermediate, low torsional barrier model is the most realistic, and that the high barrier is an overcorrection. Probably any one of the models would have had realistic enough internal coupling for this study to be made.²⁹ There is no reason to believe that our arbitrary modification of the lowest vibration frequency,¹⁶ decreasing the frequency spread of the molecule, would be a move in the non-RRKM direction.

Are we generating our effects by state selection? In the laboratory this can only be done intentionally on specialized molecules like HBC. It would be contrary to a very large amount of experience with chemical activation for us to produce similar effects with the selection procedures we have used, and even then we would expect to produce scatter around the RRKM prediction rather than unidirectional deviation. Since our phase-space procedures have no Franck-Condon principle built into them, they might well produce more uniform distributions than thermal collisions. (We are investigating this by means of trajectory studies of the collision of CH_3NC with other molecules.) Mode sampling appears to agree better with experiment than the other methods, and might be suspected on this basis of producing state distributions more similar to laboratory collisional ones. The most probable diagnosis of our results is surely Case (4) of Sec. I. B: intrinsic non-RRKM character with mild state-selection effects superposed.

Do our results arise from the use of classical

mechanics? This is a difficult question. The average energy of thermally isomerizing CH_3NC is about 40 kcal, showing extensive occupation of the vibrational ground states of the molecule. For the classical model it is around 55 kcal, not far below the 70 kcal point we calculated. The classical model has not had zero-point energy included, however, so it is not clear exactly how a comparison should be made. But it seems unlikely that $v=0$ freezing of vibrational modes will enhance intramolecular energy relaxation, which is what would be required to move in the RRKM direction.

The relevant laboratory experiments include thermal falloff measurements on CH_3NC , CH_2DNC , CD_3NC , $\text{C}_2\text{H}_5\text{NC}$, and $\text{C}_2\text{D}_5\text{NC}$. The reader is urged to study this literature carefully.^{24,30} In brief, a single type of prescription for the critical configuration is found to accommodate all the data. Critical configuration frequencies are estimated by Badger's rule and by analogy with cyclopropene, critical geometry from plausible bond order arguments. One of the external rotations is treated as active, an approximate procedure (our RRKM program uses the method of Marcus,³¹ which is much more realistic physically). The principal free parameter is the frequency of the critical configuration twist that arises from the reactant C–N–C bond. This is calibrated on CH_3NC , and the resulting value found to be satisfactory for the other molecules.

As with any rationalization procedure, if we want a basic test of the theory, we have to ask both whether it is consistent with the experiments and whether the experiments are inconsistent with its converse. If the trajectory calculations apply to the laboratory, we would suspect a false-limit kind of situation, as in Fig. 4. Thus our attention is drawn first to k_∞ , the laboratory high pressure rate constant—an energy-weighted average over the k_a measured in the computer studies. The falloff region is less interesting, because (as illustrated earlier) the decreasing collision rate interacts with RRKM and false high pressure limits in essentially the same way. Very low pressures are of no interest at all, since any consistent treatment of the collisions should give an acceptable result.

The twist frequency required for the CH_3NC k_∞ is not unbelievable. Changes on deuteration are relatively small. This is reasonable because the reactive entity is well separated from the site of isotopic (or methyl) substitution. The shape of the falloff has its rather limited isotopic variation adequately explained. These are points in favor of the orthodox interpretation of the experiments. One can only conjecture whether an adequate non-RRKM theory would make contradictory asser-

tions. (We note that at 70 kcal in Fig. 6, the relaxation treatment of our model is even further from the truth than the RRKM calculation is.)

In balancing the Monte Carlo results against the interpretations of the laboratory experiments, the reader will have to supply his own weighting factors. We anticipate that some experimentalists will think that the latter has absolute priority, and that once a series of reactions like these has been satisfyingly reconciled with the RRKM theory, it need not pass a trajectory test as well. We disagree with this. The two investigations have tested the theory in very different ways—one is purely dynamical and the other relies basically on the statistical consistency of the procedures for estimating k_a . We view them as part of an incomplete mosaic of tests. For the reasons given in the first part of the paper, we think chemical activation studies on CH_3NC and (possibly easier) $\text{C}_2\text{H}_5\text{NC}$, measurements at truly high pressures, and further theoretical work are all urgently needed.

The present impasse is no surprise. As stated earlier, we chose to study CH_3NC for historical reasons rather than because it is an ideal test case. We believe studies on other molecules will be a much better source of trajectory evidence.

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APPENDIX: ALGORITHM FOR GENERATING RANDOMLY ORIENTED n VECTORS

We start with a unit hypersphere in n dimensions, on which the final vector will end. Each step reduces the dimensionality of the problem by 1. After $i-1$ components x_j have been chosen, the problem has been reduced to that of randomly orienting a vector of length

$$r_i = \left(1 - \sum_{j=1}^{i-1} x_j^2\right)^{1/2} \quad (\text{A1})$$

in $(n-i+1)$ space. The relative probability P_i associated with the next component x_i will be the *surface* of an $(n-i)$ -dimensional hypersphere of radius

$$r_{i-1} = (r_i^2 - x_i^2)^{1/2}, \quad (\text{A2})$$

divided by the cosine of the angle between the vector with i components and the hyperplane of the $(i-1)$ -dimensional one. The angle correction is needed because of the inclination of the strip of $(n-1)$ surface bounded by the $(n-1)$ spheres for x_i and $x_i + dx_i$. The angle can also be described as

the one between x_i and vector described in Eq. (A1). These statements are necessarily somewhat clumsy in the language of a three-dimensional world. For clarification, we suggest that the user of this procedure draw his own diagram for $n=3$, $i=1$.

In k dimensions, a hypersphere of radius r has volume and surface

$$V_k = \pi^{k/2} r^k / \Gamma(\frac{1}{2}k + 1), \quad (\text{A3})$$

$$S_k = \partial V_k / \partial r = 2\pi^{k/2} r^{k-1} / \Gamma(\frac{1}{2}k). \quad (\text{A4})$$

Thus (with $k=n-i$)

$$P_i = 2\pi^{(n-i)/2} \left(1 - \sum_{j=1}^{i-1} x_j^2 - x_i^2\right)^{(n-i-1)/2} / \Gamma[\frac{1}{2}(n-i)] \cos\theta. \quad (\text{A5})$$

Since

$$\cos\theta = \left[\left(1 - \sum_{j=1}^{i-1} x_j^2 - x_i^2\right) / \left(1 - \sum_{j=1}^{i-1} x_j^2\right) \right]^{1/2}, \quad (\text{A6})$$

the relative probability, shorn of numerical constants, becomes

$$P_i = \left(1 - \sum_{j=1}^{i-1} x_j^2 - x_i^2\right)^{(n-i-2)/2}. \quad (\text{A7})$$

Here we have taken note of the constancy of the denominator in Eq. (A6) throughout the i th selection step. The most probable value of x_i is 0, for which

$$P_i^0 = \left(1 - \sum_{j=1}^{i-1} x_j^2\right)^{(n-i-2)/2}. \quad (\text{A8})$$

Conventional rejection methods can be conveniently used to sample Eq. (A7).

The computational procedure can be arranged as follows. Let R denote a freshly generated random number, $0 < R < 1$. Initially $i=1$ and the sums in Eqs. (A7) and (A8) are 0. Randomly generate x_1 , $0 < x_1 < 1$. Accept it if

$$(P_1/P_1^0) > R. \quad (\text{A9})$$

Otherwise try a new x_1 and R . [The R in Eq. (A9) and the one that generates x_1 are different.] Continue until an x_1 is accepted. On subsequent steps the procedure is the same except that the sums are nonzero and the sampling range of x_i is

$$0 < x_i < r_i, \quad (\text{A10})$$

with r_i given by Eq. (A1).

When i reaches $n-1$ the procedure changes, since 0 is no longer the most probable value of x_i . But the relative probability has become integrable, and direct sampling can now be used:

$$P_{n-1} = \left(1 - \sum_{j=1}^{n-2} x_j^2 - x_{n-1}^2\right)^{-1/2}, \quad (\text{A11})$$

$$c(x) = \int_0^x P_{n-1} dx_{n-1} = \sin^{-1}(x/r_{n-1}), \quad (\text{A12})$$

$$c(x_{n-1})/c(r_{n-1}) = R = (2/\pi)\sin^{-1}(x_{n-1}/r_{n-1}), \quad (\text{A13})$$

$$x_{n-1} = \left(1 - \sum_{j=1}^{n-2} x_j^2\right) \sin(\frac{1}{2}\pi R). \quad (\text{A14})$$

Finally,

$$x_n = \left(1 - \sum_{j=1}^{n-1} x_j^2\right)^{1/2}. \quad (\text{A15})$$

It is convenient to let $x_i > 0$ throughout the selection procedure, and randomize signs afterward. The results are very easily tested by generating a number of vectors and comparing the distributions of x_1 and x_n . Any error will be amplified into a severe discrepancy between these.

*Address reprint requests to D.L.B.

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